A Two-Centre Shield Law for Twin-Prime Gaps on the $6N$ Skeleton: Modular-Shift Gain and a Cross-Prime Hedge

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Abstract

Part IV closed the long-gap (210) residual of the conditional twin-gap singular series with a single-centre spatial-compression penalty, but left the short-gap (42) residual, and posed the genuine two-centre correlation between the factorisations of N and $N+d$ as the open problem. Here we measure that correlation directly and obtain a quantitative law. Conditioning a twin centre N on divisibility by a prime q, we measure the shield $S(d,q) = P(N+d\,\text{twin} \mid q \mid N)/P(N+d\,\text{twin})$ on the 23,988,173 twin centres of S_{10}. We find $S(d,q)$ factors cleanly into two measured pieces: a *modular-shift gain* (because $q \mid N$ forces $N+d \equiv d \pmod q$, the right centre's q-survival jumps from its real baseline rate to certainty when d is q-safe) and a *cross-prime hedge* (the same conditioning shifts the right centre's residue distribution at the other primes, mainly $q' = 5$, changing their survival). The product $S(d,q) = \text{gain} \times \text{hedge}$ reproduces the measured shield for both $6\Delta N = 42$ and 210, across all admitting primes, to within 1.5%, stable between S_9 and S_{10}. The hedge is below 1 for 42 and above 1 for 210, which is why the single-centre theory of Part IV under- and over-shot respectively. Both pieces require the *measured* modular distribution of twin centres (which carries the Part I enrichment); an idealised uniform assumption fails. We make no claim about the infinitude of twin primes or any k-tuple conjecture, and we do not yet bridge this right-centre law back to the gap-preference distribution of Part IV, which we keep as the open problem.

1 The object: a measured two-centre shield

Parts II–IV studied $r(d \mid \omega)$, the preference for gap $\Delta N = d$ among twin centres of factor count ω. Part IV reduced its residual to a single quantity it could not model from N alone: the correlation between the factorisations of the two centres N and $N+d$. We isolate that correlation in its cleanest form.

Definition 1 (two-centre shield)**.** *For a centre-step d and a prime $q > 3$,*

$$S(d,q) \;=\; \frac{P\big(N+d \textit{ is a twin centre} \mid q \mid N,\ N \textit{ twin}\big)}{P\big(N+d \textit{ is a twin centre} \mid N \textit{ twin}\big)},$$

measured over a high-ω band ($\omega_{>3} \in \{4,5,6\}$) of S_{10}.

$S > 1$ means dividing the centre by q raises the chance that the next skeleton position d steps away is also a twin centre. The interferometric measurement (Table 1) shows S is large and structured: for $6\Delta N = 42$ it is dominated by $q = 5$ (not by $q = 7$, the prime dividing the gap, contrary to a natural guess), and for 210 it rises across the mid-range primes.

2 The law: modular-shift gain times cross-prime hedge

Two measured factors reproduce $S(d,q)$.

Modular-shift gain. A twin centre needs $N \not\equiv \pm 6^{-1} \pmod q$ for every q; write $\mathrm{dead}(q) = \{\pm 6^{-1} \bmod q\}$. If $q \mid N$ then $N \equiv 0$, so the right centre sits at $N + d \equiv d \pmod q$, a *fixed* residue. When $d \notin \mathrm{dead}(q)$ the right centre is then certain to be q-safe, versus its baseline q-safety rate $\beta_q(d) = P(N{+}d\ q\text{-safe} \mid N \text{ twin})$ measured over all twins. The gain is

$$g(d,q) = \frac{\mathbf{1}\{d \notin \mathrm{dead}(q)\}}{\beta_q(d)}.$$

The baseline $\beta_q(d)$ must be the *measured* rate (it inherits the Part I twin-centre bias modulo q); the idealised uniform value overstates the gain (e.g. for $q = 5, d = 7$ uniform gives 1.5, measured gives 1.236).

Cross-prime hedge. Conditioning on $q \mid N$ also moves the right centre's residue distribution at every *other* prime q', changing its q'-safety. The hedge is the measured product

$$h(d,q) = \prod_{q' \neq q} \frac{P(N{+}d\ q'\text{-safe} \mid q \mid N)}{\beta_{q'}(d)}.$$

Empirically the hedge is carried almost entirely by $q' = 5$ (the smallest odd admissible prime, with the largest dead fraction): conditioning on any $q \mid N$ slightly changes how often the right centre lands in dead(5).

Proposition 1 (two-centre shield law)**.** *$S(d,q) = g(d,q)\,h(d,q)$, with both factors evaluated from the measured modular distribution of twin centres.*

3 Confrontation with S_{10}

Table 1 and Fig. 1 test Proposition 1 on the 23,988,173 twin centres of S_{10}.

	$6\Delta N = 42$				$6\Delta N = 210$			
q	gain	hedge	pred.	meas.	gain	hedge	pred.	meas.
5	1.236	0.994	1.229	1.229	1.000	1.060	1.060	1.060
7	1.000	0.946	0.946	0.943	1.000	1.083	1.083	1.079
13	1.190	0.945	1.124	1.118	1.087	1.103	1.199	1.192
17	1.138	0.938	1.068	1.065	1.138	1.108	1.261	1.266
23	1.098	0.933	1.024	1.025	1.098	1.105	1.213	1.230
29	1.076	0.932	1.003	1.012	1.076	1.101	1.185	1.167
31	1.071	0.933	0.999	0.996	1.071	1.102	1.179	1.180
37	1.058	0.933	0.988	0.984	1.058	1.099	1.163	1.156

Table 1: Two-centre shield law versus measurement, S_{10}. Maximum error 1.0% (42) and 1.5% (210). Primes dividing the gap $(7, 11, 19, 41, 43)$ are omitted where they lock the gap outright (shield = 0). The hedge is systematically < 1 for 42 and > 1 for 210.

The law closes to $\leq 1.5\%$ at every admitting prime, for both gaps, and the same closure holds in S_9 (maximum error 3% on a $64\times$ smaller sample). Two qualitative facts fall out. First, the 42

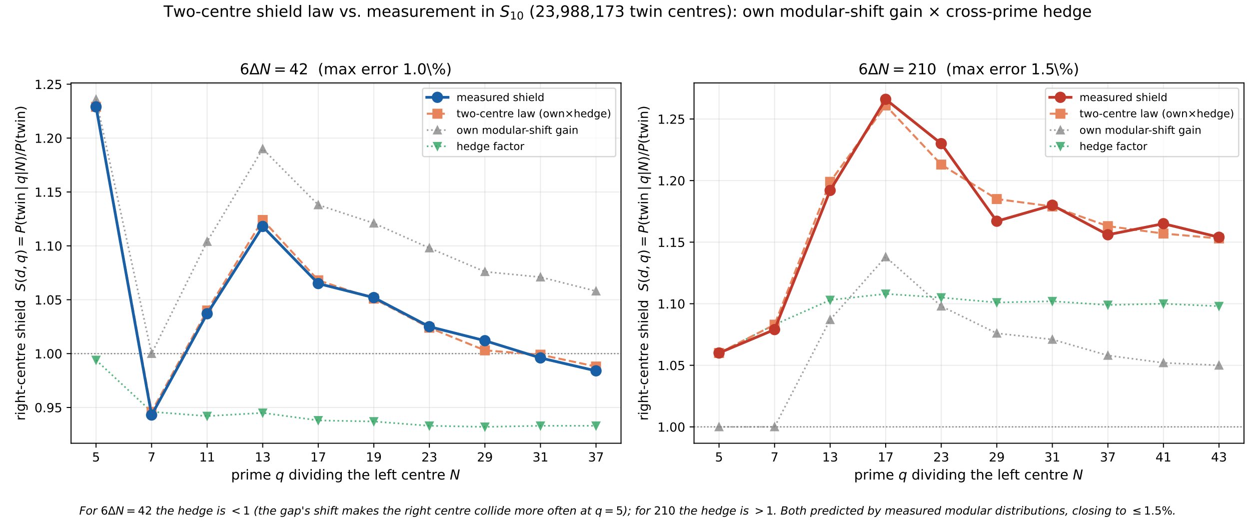

Figure 1: Measured shield $S(d, q)$ (solid) versus the law $g \times h$ (dashed), with the two factors shown separately. The gain (grey) and hedge (green) must be multiplied to match; the hedge is < 1 for 42 and > 1 for 210.

shield is a $q = 5$ effect: 42 has $\Delta N = 7$, and $N + 7 \equiv N + 2 \pmod 5$, so $5 \mid N$ places the right centre at the safe residue 2; the prime 7 that divides the physical gap is nearly inert ($g(42, 7) = 1$). This corrects the natural "$7 \mid 42$ is the shield" guess. Second, the hedge sign differs by gap: for 42 the right centre's shift makes it *more* likely to collide at 5 (hedge < 1), for 210 *less* likely (hedge > 1). This is exactly why the single-centre model of Part IV undershot 42 and the absolute survival of 210 behaved as it did.

4 What is not yet done: the bridge to the gap distribution

Proposition 1 is a law for the *right-centre conditional survival* given the left centre's factors. It is not yet the gap-preference $r(d \mid \omega)$ of Part IV. Passing from one to the other requires summing the shield over the factor sets that constitute a stratum ω and integrating the right-centre survival into a normalised gap distribution—a step we have not carried out and which need not be a simple average, since the shield depends on *which* primes divide N, not only on how many.

> **Open problem.** Bridge the two-centre shield law $S(d, q) = g\,h$ to the stratified gap preference $r(d \mid \omega)$ of Part IV, and determine whether it accounts for the residual high-ω rise of 42 (observed 1.55 at $\omega = 6$ in S_{10}). The shield is established at the level of single conditioning primes and their measured cross-prime hedge; the bridge to the ω-stratified distribution is open.

5 Conclusion

The two-centre correlation that Parts II–IV isolated as the missing mechanism is, at the level of the right-centre conditional survival, a clean and measurable product of a modular-shift gain and a cross-prime hedge, closing to $\leq 1.5\%$ on 2.4×10^7 twin centres. Both factors are genuinely two-centre and both require the measured (not idealised) modular distribution of twin centres. The mechanism overturns two natural guesses—that the gap's own prime factor (7 for 42) is the shield, and that cross-prime effects are negligible—and shows instead that the shield is a $q = 5$ modular-shift effect dressed by a small, sign-definite hedge. We do not bridge this back to the gap-preference distribution here; that, and with it the final fate of the 42 residual, is left as the open problem. As throughout, no claim is made about the infinitude of twin primes or any k-tuple conjecture; this is a measured, factor-resolved law for the conditional structure of the $6N$ twin skeleton.

Data and code availability. The interferometric scan and the gain/hedge decomposition are openly available [6]; the S_{10} twin count 23,988,173 matches Part I.

References

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